TRINITY WESTERN UNIVERSITY

# CMPT 461 Immersive Computing

Assignment #6: VR Interaction Fundamentals (C# Scripting & Physics)

Due Date: March 2nd, 2026 (by 11:59 PM)

**Purpose:**

Students build the core interaction mechanics they will reuse in their team VR application.

## 🎯 Learning Outcomes

Students will learn to:

- Create and attach C# scripts in Unity
- Use object references safely
- Configure colliders, rigidbodies, triggers
- Implement basic XR interactions (grab, button, socket, teleport)
- Build reusable prefabs

## Assignment Description

### 📚 Part 0 — XR Interaction Toolkit Setup (Step-by-Step)

**Tutorial**

https://www.youtube.com/watch?v=rKDuYzS-D2E
https://www.youtube.com/watch?v=zbqHNwDpi6Y

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**Step 0.1 — Create Project**

1. Open Unity Hub
2. New Project → 3D (URP recommended but optional)
3. Name project VR_Interaction_Assignment

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**Step 0.2 — Install XR Interaction Toolkit**

1. Window → Package Manager
2. Select **Unity Registry**
3. Search **XR Interaction Toolkit**
4. Click Install
5. When prompted → Enable Input System → restart Unity

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**Step 0.3 — Import Samples (important)**

1. Select XR Interaction Toolkit in Package Manager
2. Samples tab → Import:
    - Starter Assets
    - Default Input Actions

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**Step 0.4 — Add XR Origin**

1. GameObject → XR → XR Origin (Action-based)
2. Delete Main Camera (XR Origin contains camera)

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**Step 0.5 — Add Teleport Area**

1. Create Plane (floor)
2. Add component:
    - Teleportation Area

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**Step 0.6 — Test Scene**

Press Play:

- Controllers visible
- Teleport works

✅ Deliverable: screenshots of the teleportation area and demo video(s)

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## 📚 Part A — C# Foundations (Very Guided)

**Tutorial**

https://www.youtube.com/watch?v=pIDvwImZvW8

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**Task A1 — Toggle Object Script (Detailed)**

**Goal**

Press a key → object appears/disappears.

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**Steps**

1. Create Cube → name ToggleCube
2. Create folder Scripts
3. Right-click → Create → C# Script → ToggleObject
4. Double-click script

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**Minimal script structure students should write**

- Create public GameObject variable
- Inside Update()
- Detect key press
- Toggle SetActive

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**Back in Unity**

1. Drag script onto empty GameObject called Manager
2. Drag ToggleCube into script field
3. Press Play → test

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**Task A2 — Material Change Script**
**Goal**
Interaction → object changes color.

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**Steps**

1. Create script ChangeMaterial
2. Add:
    - Renderer reference
    - Two materials
3. On interaction:
    - Swap material

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**Unity Steps**

1. Create two materials (red, blue)
2. Attach script to cube
3. Assign materials
4. Test

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**Task A3 — Timer / State Script**
**Goal**
After 3 seconds → action occurs.

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**Steps**

1. Create script TimerAction
2. Add float timer
3. Increase timer using Time.deltaTime
4. If timer > value → trigger action

Example actions:

- enable object
- change color
- play sound

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✅ Deliverable: 3 scripts + demo video

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## 🟩🟦 Part B — Physics & Events (Step-by-Step)

**Tutorials**
https://www.youtube.com/watch?v=C_bOx2WEORM
https://www.youtube.com/watch?v=Au8w_6qZxT8

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**Task B1 — Trigger Zone Event**
**Goal**

Player enters zone → event happens.

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**Steps**

1. Create Cube → name TriggerZone
2. Add Box Collider → check **Is Trigger**
3. Create script TriggerEvent
4. Use OnTriggerEnter

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**Test Event Ideas**

- Turn on light
- Spawn object
- Change color

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**Task B2 — Throwable Object (Very Important)**

**Steps**

1. Create object (sphere recommended)
2. Add:
    - Rigidbody
    - XR Grab Interactable

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**Important Settings (students often miss)**

Rigidbody:

- Use Gravity ON
- Mass ~1
- Collision detection → Continuous

XR Grab Interactable:

- Movement type → Velocity Tracking

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**Test**

Grab → throw → observe physics

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**Task B3 — Collision Feedback**

**Steps**

1. Create script CollisionFeedback
2. Use OnCollisionEnter
3. Trigger feedback:
    - sound
    - color
    - particle

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✅ Deliverable: screenshots of physics playground scene and demo video(s) of Tasks B1, B2, and B3

## 📚 Part C — Core VR Interaction Scene (Main Task — Highly Guided)

**Tutorial**
https://www.youtube.com/watch?v=PhIyTqgIjfA

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**Requirement 1 — Grabbable Object**
Steps:

1. Create object
2. Add Rigidbody
3. Add XR Grab Interactable
4. Test grab

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**Requirement 2 — Button / Lever Interaction**
Simplest version (recommended for struggling students):
Steps:

1. Create cube as button
2. Add XR Simple Interactable
3. Create script ButtonAction
4. Use Select Entered event
5. Trigger action (door open, light toggle)

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**Requirement 3 — Socket Interaction (Important)**
Steps:

1. Create empty object → Add XR Socket Interactor
2. Create object with XR Grab Interactable
3. Place object into socket → event fires

Example event:

- puzzle solved
- door opens
- UI appears

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**Requirement 4 — Teleport Area**
Steps:

1. Create floor
2. Add Teleportation Area
3. Ensure locomotion system exists
4. Test teleport

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**Scene Requirement (Simple Explanation)**
Create a small VR experience that includes:

- 3 interactions minimum
- 1 physics object

- 1 interaction chain

Example (very beginner friendly):

- Grab key → place in socket → door opens

✅ Deliverable: demo video(s) of Requirements 1, 2, 3, and 4, and Scene Requirement

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⭐ **Extra Support Advice (Include in Assignment)**

If something does not work:

1. Check Console errors first
2. Verify references are assigned in Inspector
3. Confirm Rigidbody exists for physics interactions
4. Confirm collider settings (Is Trigger vs not)
5. Test interaction in Play mode slowly

**Marking scheme**:

- Part 0 (10 marks): screenshots of the teleportation area and demo video(s)
- Part A (10 marks): 3 scripts (ToggleObject, ChangeMaterial, and TimerAction) and demo video(s) of Tasks A1, A2, and A3
- Part B (10 marks): Scripts and screenshots of physics playground scene and demo video(s) of Tasks B1, B2, and B3
- Part C (15 marks): demo video(s) of Requirements 1, 2, 3, and 4, and Scene Requirement
- Writing a short reflection on (5 marks):
    - What interaction will be reused in your team project
    - Biggest difficulty

Total marks in Assignment #5: 50 marks.

**Assignment Submission:**

You need to submit the following to Moodle before the due date:

- A document (PDF preferred) with your reflections and screenshots of Part 0 and B
- A zip file of your Unity project **standalone executable** and its necessary folder(s) and files for Parts 0, A, B, and C, including DLLs to run and play your assignment (not your original Unity project)
- Demo video files of screen recordings

**Note:** There are instructions on how to create a Unity project standalone executable in the previous assignment instructions.

When your Unity project **standalone executable** is executed on the computers of the instructor or TA, it should be playable without any errors. The instructor or TA will not fix any errors to evaluate your assignment.

Note: If your file size is greater than 20 MB, upload your files (or a zip file) to an online storage service, such as OneDrive, Google Drive, or Dropbox, and submit the download link. Please make sure that anybody can download from the link.